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1. Derive each of the convolutions listed in Table 2-2 of your textbook, and submit your work. Item 1 is worth 10 points, and the rest are worth 15 points each.

**Table 2-2:** Commonly encountered convolutions.

1.  $u(t) * u(t) = t u(t)$

2.  $e^{at} u(t) * u(t) = \left( \frac{e^{at} - 1}{a} \right) u(t)$

3.  $e^{at} u(t) * e^{bt} u(t) = \left[ \frac{e^{at} - e^{bt}}{a - b} \right] u(t), \quad a \neq b$

4.  $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$

5.  $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at} + (a - b) t e^{at}}{(a - b)^2} u(t),$   
 $a \neq b$

6.  $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$

7.  $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$

$$1. \quad u(t) * u(t) = t u(t)$$

$$\int_{-\infty}^{\infty} u(z) u(t-z) dz$$

$$z > 0$$

$$t > z, \text{ so } 0 < z < t$$

$$= u(t) \int_0^t 1 dz = u(t) \left[ z \right]_0^t = t u(t)$$

$$2. \quad e^{at} u(t) * u(t) = \left( \frac{e^{at} - 1}{a} \right) u(t)$$

$$y(t) = \int_{-\infty}^{\infty} e^{az} u(z) \cdot u(t-z) dz$$

$$z > 0$$

$$t > 0, \text{ so } 0 < z < t$$

$$\Rightarrow y(t) = u(t) \int_0^t e^{az} dz$$

$$= u(t) \left[ \frac{e^{az}}{a} \right]_0^t = u(t) \left( \frac{e^{at} - 1}{a} \right)$$

$$3. \quad e^{at} u(t) * e^{bt} u(t) = \left[ \frac{e^{at} - e^{bt}}{a - b} \right] u(t), \quad a \neq b$$

$$\int_{-\infty}^{\infty} e^{az} u(z) e^{b(t-z)} \cdot u(t-z) dz$$

$$z > 0$$

$$t > z, \text{ so } 0 < z < t$$

$$\Rightarrow u(t) \int_0^t e^{az} e^{b(t-z)} dz \Rightarrow u(t) \int_0^t e^{az} e^{bt} \cdot e^{-bz} dz$$

# Worksheet 4 cont. 1

3.

$$\begin{aligned}
 y(t) &= u(t) \int_0^t e^{az} e^{bt} \cdot e^{-bz} dz \\
 &= u(t) e^{bt} \int_0^t e^{az-bz} dz \\
 &= u(t) e^{bt} \int_0^t z(a-b) dz \\
 &= u(t) e^{bt} \left( \frac{z(a-b)}{a-b} \Big|_0^t \right) = u(t) e^{bt} \left( \frac{e^{t(a-b)} - 1}{a-b} \right) \\
 &= u(t) \frac{e^{at} - e^{bt}}{a-b} \\
 &= u(t) \left( \frac{e^{at} - e^{bt}}{a-b} \right)
 \end{aligned}$$

4.  $e^{at} u(t) * e^{at} u(t) = te^{at} u(t)$

$$\int_{-\infty}^{\infty} e^{az} u(z) e^{a(t-z)} u(t-z) dz$$

$z > 0$  and  $t > 0$ , so  $0 < z < t$

$$\begin{aligned}
 \Rightarrow u(t) \int_0^t e^{az} e^{a(t-z)} dz &= u(t) e^{at} \int_0^t 1 dz \\
 &= u(t) e^{at} \cdot t
 \end{aligned}$$

# Worksheet 4 cont. 2

5.  $te^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at} + (a-b)te^{at}}{(a-b)^2} u(t),$   
 $a \neq b$

$$y(t) = \int_{-\infty}^{\infty} ze^{az} u(z) e^{b(t-z)} u(t-z) dz$$

Commutativity  
 $\Rightarrow x(t) * h(t) = h(t) * x(t)$

$z \geq 0$  and  $t \geq z$ , so  $0 \leq z \leq t$

$$\Rightarrow y(t) = u(t) \int_0^t ze^{az} e^{bt-bz} dz$$

$$= u(t) e^{bt} \int_0^t ze^{z(a-b)} dz \quad \leftarrow \text{IBP}$$

$$uv - \int v dv$$

$$v = z$$

$$dv = dz$$

$$\frac{dv}{v} = \frac{z(a-b)}{a-b} dz$$

$$= u(t) e^{bt} \left[ \frac{z \cdot e^{z(a-b)}}{a-b} \Big|_0^t - \int_0^t \frac{e^{z(a-b)}}{a-b} dz \right]$$

$$= u(t) e^{bt} \left[ \frac{z e^{z(a-b)}}{a-b} \Big|_0^t - \frac{e^{z(a-b)}}{(a-b)^2} \Big|_0^t \right]$$

$$= u(t) e^{bt} \left[ \left( \frac{t e^{t(a-b)}}{a-b} - 0 \right) - \left( \frac{e^{t(a-b)}}{(a-b)^2} - \frac{1}{(a-b)^2} \right) \right]$$

$$= u(t) e^{bt} \left[ \frac{t e^{t(a-b)} - \frac{e^{t(a-b)}}{(a-b)} + \frac{1}{(a-b)^2}}{(a-b)^2} \right]$$

$$= u(t) e^{bt} \left[ \frac{t e^{at} (a-b) - e^{at} + 1}{(a-b)^2} \right]$$

$$= u(t) \left( \frac{t e^{at} (a-b) - e^{at} + e^{bt}}{(a-b)^2} \right)$$

$$= u(t) \left( \frac{t e^{at} (a-b) - e^{at} + e^{bt}}{(a-b)^2} \right)$$

# Worksheet 4. cont. 3

6.  $\mathcal{L}^{-1}\{t e^{at} u(t)\} * \mathcal{L}^{-1}\{e^{at} u(t)\} = \frac{1}{2} t^2 e^{at} u(t)$   $x(t) * u(t) = u(t) * x(t)$

$$y(t) = \int_{-\infty}^{\infty} z e^{az} u(z) e^{a(t-z)} u(t-z) dz$$

$z \geq 0$  and  $t \geq z$ , so  $0 \leq z \leq t$

$$= u(t) \int_0^t z e^{az} e^{a(t-z)} dz$$

$$= u(t) e^{at} \int_0^t z dz = u(t) e^{at} \left( \frac{z^2}{2} \Big|_0^t \right) = u(t) \frac{e^{at} t^2}{2}$$

7.  $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$

$$y(t) = \int_{-\infty}^{\infty} \delta(z - T_1) \cdot \delta(t - z - T_2) dz$$

$$= \int_{-\infty}^{\infty} \delta(T_1) \cdot \delta(t - z - T_2) dz$$

Remember sampling property  
 $\Rightarrow x(t) \delta(t) = x(0) \delta(t)$   
 $\Rightarrow x(t) \delta(t - T) = x(T) \delta(t - T)$

$\Rightarrow$  set  $z = T_1$

$$\text{so } \delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$$